

ARK-448 Run Log

Experiment: ARK-448 — Dynamical Decoupling vs. Baseline Under an Idle Window

Backend: ibm_marrakesh (IBM Quantum open plan)

Date: 2026-07-17

Final outcome: ABORTED AT SPAM GATE (honest gate-stop)

Chronology

1. **LOCK.** Preregistration and all analysis code committed on branch `execute/ark-448` (branched from `origin/main` @ `bb3cf66`, which includes merged ARK-447). Commit `e947b1c`, tag `ark-448-v1.0-lock`. `MANIFEST.txt` records SHA-256 of the 7 locked files.
2. **Offline validation.** Full pipeline validated with a noiseless Aer simulation before any hardware submission; all simulator stub files were deleted so no simulated data was ever committed as hardware data.
3. **Budget check.** IBM open-plan budget confirmed limited (~186 s remaining pre-run). Backends `ibm_fez`, `ibm_marrakesh`, `ibm_kingston` all operational.
4. **Qubit selection.** `ark_448_select_qubits.py` → `Q_A=1`, `Q_P=2` on `ibm_marrakesh` (sum readout error 0.00537, lowest connected pair; series-consistent with ARK-447).
→ `selected_qubits.json`, `calibration_snapshot_ibm_marrakesh_20260717.json`.
5. **SPAM gate.** `ark_448_spam_job.py` submitted (job `d9cqji4inv1c73ao54eg`, 2048 shots/circuit).
Result:
- SPAM_A error = 0.01025 (≤ 0.02) → **PASS**
- SPAM_P $|P(1)-0.5| = 0.02197$ (≤ 0.02 required) → **FAIL** (over by 0.00197, $\approx 2.0\sigma$ at 2048 shots)
- **gate_passed = false**
→ `spam_results.json`, `spam_run.log`.
6. **HALT per protocol.** Because the SPAM gate failed, the principal 4-arm job was **not** submitted. No `raw_results.json`, no `principal_job_*` files exist. The LOCKED preregistration does not permit re-running the gate or loosening the threshold, so execution stopped.
7. **Gate-stop record.** `proofrecord.json` written documenting the abort (real SPAM data only, explicit no-fabrication statement, no principal data). `RESULTS.md`, `README.md`, and `MANIFEST_v1.1.txt` added.

Budget accounting

- Post-run IBM budget check: **~183 s remaining, ~417 s consumed** for the session.
- The SPAM job used minimal budget; the principal job never ran, conserving remaining seconds.
- No shot-reduction contingency (Section 4) was triggered because the run never reached the principal job.

Deviations from preregistration

None. The abort is an expected, preregistered branch (SPAM gate must pass before principal job).
No criteria were altered.